

# Internet of Things

By

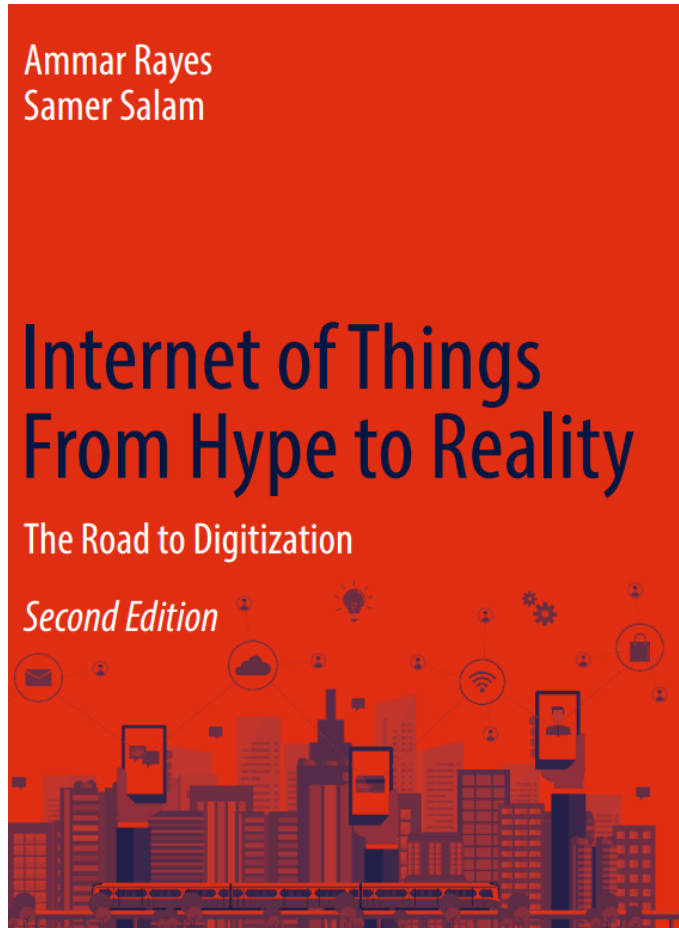
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# Reference book



## Chapter 8

# Outline

- ▶ Internet of Things Security and Privacy.
- ▶ Introduction of IoT security and privacy
- ▶ IoT Security Challenges
- ▶ IoT Security Requirements
- ▶ IoT Three-Domain Architecture
- ▶ Cloud Domain Attacks and Countermeasures
- ▶ Fog Domain Attacks and Countermeasures
- ▶ Sensing Domain Attacks and Countermeasures
- ▶ Summary

# Introduction

- ▶ The Internet of Things (IoT) promises to make our lives more convenient by turning each physical object in our surrounding environment into a smart object that can sense the environment, communicate with the remaining smart objects, perform reasoning, and respond properly to changes in the surrounding environment.
- ▶ However, IoT brings are also associated with new security risks and privacy issues that must be addressed properly.
- ▶ Ignoring these security and privacy issues will have serious effects on the different aspects of our lives including the homes we live in, the cars we ride to work, and even the effects that will reach our own bodies.

# Introduction

- ▶ smart camera that notifies your smartphone during business hours when movement is detected, the smart door that opens remotely, and the smart fridge that notifies you when you are short of milk. Imagine now the level of control that an attacker can gain by hacking those smart meters and gadgets if the security of those devices was overlooked.
- ▶ In fact, the damage caused by cyberattacks in the IoT era will have a direct impact on all the physical objects that you use in your daily life.
- ▶ The same applies to your smart car as the number of integrated sensors continues to grow rapidly and as the wireless control capabilities increase significantly over time, giving an attacker who hacks the car the ability to control the windshield wipers, the radio, the door lock, and even the brakes and the steering wheel of your car.

# Introduction

- ▶ Maintaining users' privacy in IoT is also crucial as there is an enormous amount of information that an outsider can learn about people's life by **eavesdropping** on the sensed data that their smart house appliances and wearable devices report.
- ▶ Even if your IoT objects are merely reporting metadata, you would be surprised by the amount of information that an outsider can learn about your personal life when aggregating the metadata collected from multiple hacked objects that surround you over time.

# Device-Level Threats

- ▶ **Weak or Default Passwords:** Many IoT devices ship with default or easily guessable passwords, making them simple targets for brute-force attacks and unauthorized access.
- ▶ **Insecure Firmware:** Outdated or vulnerable firmware can contain known security flaws that attackers can exploit. Lack of secure update mechanisms also prevents patching these vulnerabilities.
- ▶ **Hardware Vulnerabilities:** Flaws in the hardware itself, such as insecure bootloaders or the lack of physical security, can be exploited to gain control of the device.
- ▶ **Insecure Storage:** Sensitive data stored on the device without proper encryption can be easily accessed if the device is compromised.
- ▶ **Lack of Encryption:** Transmitting data without encryption allows attackers to intercept and read sensitive information.
- ▶ **Physical Tampering:** Devices deployed in accessible locations can be physically manipulated to gain unauthorized access or cause damage.

# Network-Level Threats

- ▶ **Network Sniffing and Eavesdropping:** Attackers can intercept unencrypted network traffic to steal credentials or sensitive data.
- ▶ **Man-in-the-Middle (MitM) Attacks:** Attackers can intercept communication between devices and servers, potentially altering data or injecting malicious commands.
- ▶ **Denial-of-Service (DoS) and Distributed Denial-of-Service (DDoS) Attacks:** Compromised IoT devices can be used as part of a botnet to flood target systems with traffic, making them unavailable. The Mirai botnet is a notorious example of this.
- ▶ **Compromised Network Infrastructure:** Vulnerabilities in routers, gateways, and other network devices can provide attackers with access to the entire IoT network



# Cloud-Level Threats

- ▶ **Insecure Cloud Interfaces:** Weak APIs or insecure web interfaces to cloud platforms can be exploited to gain unauthorized access to device data and control functions.
- ▶ **Data Breaches in the Cloud:** Vulnerabilities in cloud storage or processing systems can lead to the exposure of large amounts of sensitive IoT data.
- ▶ **Account Hijacking:** Attackers can gain control of user accounts associated with IoT devices and cloud services.
- ▶ **Insecure Ecosystem Interfaces:** Weak security in the interfaces between different components of the IoT ecosystem
- ▶ **Supply Chain Vulnerabilities:** Compromised components or software introduced during the manufacturing or distribution process can create backdoors or vulnerabilities in IoT devices.

# IoT Security Challenges

- (a) **Multiple Technologies:** IoT combines multiple technologies such as radiofrequency identification (RFID), wireless sensor networks, cloud computing, virtualization, etc. Each of these technologies has its own vulnerabilities.
- (b) **Multiple Verticals:** The IoT paradigm will have numerous applications (also called verticals) that span eHealth, industrial, smart home gadgets, smart cities, etc. The security requirements of each vertical are quite different from the remaining verticals.
- (c) **Scalability:** None of the previously proposed centralized defensive frameworks can work anymore with the IoT paradigm, where the focus must be switched to finding practical decentralized defensive security mechanisms.
- (d) **Availability:** Security plays a major rule in high availability as network administrators often hesitate to use needed threat-response technology functions for fear that such functions will take down critical systems.

# IoT Security Challenges

- ▶ **Big Data:** Not only the number of smart objects will be huge, but also the data generated by each object will be enormous as each smart object is expected to be supplied by numerous sensors, where each sensor generates huge streams of data over time. This makes it essential to come up with efficient defensive mechanisms that can secure these large streams of data.
- ▶ **Resource Limitations:** The majority of IoT end devices have limited resource capabilities such as CPU, memory, storage, battery, and transmission range. This makes those devices a low-hanging-fruit for denial of service (DoS) attacks where the attacker can easily overwhelm the limited resource capabilities of those devices causing a service disruption
- ▶ **Remote Locations:** In many IoT verticals (e.g., smart grid, railways, roadsides), IoT devices, especially sensors, will be installed in unmanned locations that are difficult to reach. Attackers can interfere with these devices without being seen.

# IoT Security Challenges

- ▶ **Mobility:** Smart objects are expected to change their location often in the IoT paradigm. This adds extra difficulties when developing efficient defensive mechanisms in such dynamic environments.
- ▶ **Delay-Sensitive Service:** The majority of IoT applications are expected to be delay-sensitive, and thus one should protect the different IoT components from any attack that may degrade their service time or may cause a service disruption.

# IoT Security Requirements

- ▶ These requirements include:
- ▶ **Confidentiality:** ensures that the exchanged messages can be understood only by the intended entities.
- ▶ **Integrity:** ensures that the exchanged messages were not altered/tampered by a third party.
- ▶ **Authentication:** ensures that the entities involved in any operation are who they claim to be. A masquerade attack or an impersonation attack usually targets this requirement where an entity claims to be another identity.
- ▶ **Availability:** ensures that the service is not interrupted. Denial of service attacks target this requirement as they cause service disruption.

# IoT Security Requirements

- ▶ **Authorization:** ensures that entities have the required control permissions to perform the operation they request to perform.
- ▶ **Freshness:** ensures that the data is fresh. Replay attacks target this requirement where an old message is replayed in order to return an entity into an old state.
- ▶ **Non-repudiation:** ensures that an entity can't deny an action that it has performed.
- ▶ **Forward Secrecy:** ensures that when an object leaves the network, it will not understand the communications that are exchanged after its departure.
- ▶ **Backward Secrecy:** ensures that any new object that joins the network will not be able to understand the communications that were exchanged prior to joining the network.

# IoT Three-Domain Architecture

- ▶ The three domain architecture that we consider in our security analysis
- ▶ (a) **IoT Sensing Domain**: This domain is made up of all the smart objects that have the capability to sense the surrounding environment and report the sensed data to one of the devices in the fog domain. The smart objects in the sensing domain are expected to change their location over time.

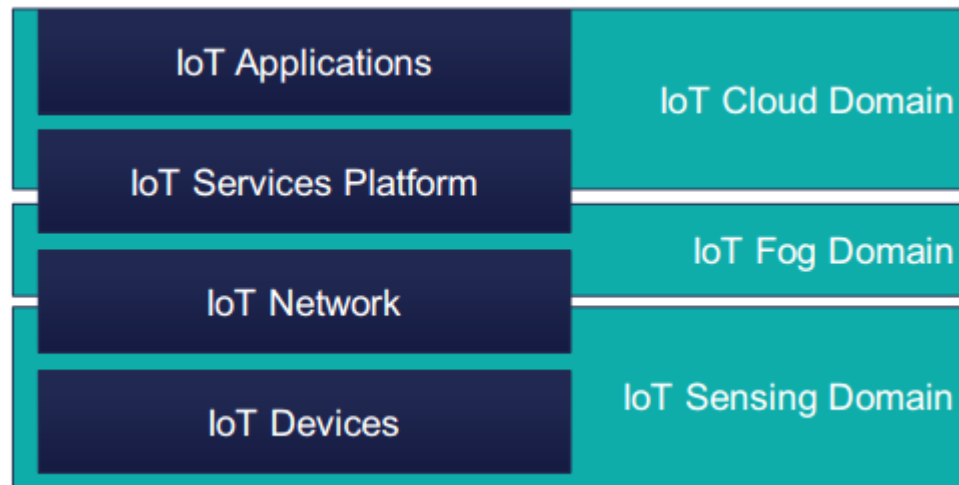


Fig. 8.1 Mapping of IoT domains

# IoT Three-Domain Architecture

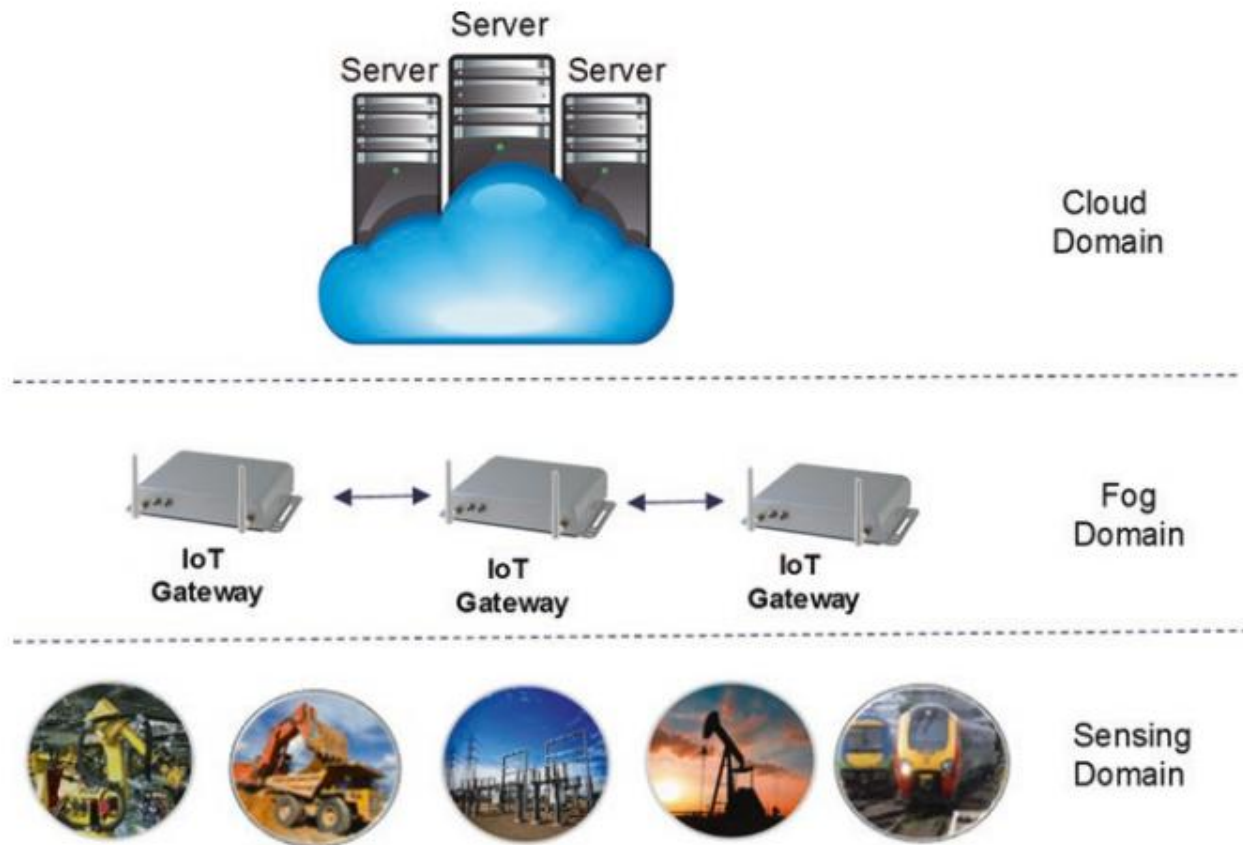


Fig. 8.2 The IoT domains



# IoT Three-Domain Architecture

- ▶ (b) **Fog Domain:** This domain consists of a set of fog devices that are located in areas that are highly populated by many smart objects. Each fog device is allocated a set of smart objects where the allocated objects report their sensed data to the fog device. The fog device performs operations on the collected data including aggregation, preprocessing, and storage. Fog devices are also connected with each other in order to manage the communication among the smart objects and in order to coordinate which fog device will be responsible for handling which object as objects change their location over time. Each fog device is also connected to one or multiple servers in the cloud domain.
- ▶ (c) **Cloud Domain:** This domain is composed of a large number of servers that host the applications that are responsible for performing the heavy-computational processing operations on the data reported from the fog devices.

# Cloud Domain Attacks and Counter measures

- ▶ Each IoT application is dedicated one or multiple virtual machines (VMs) where each VM is assigned to one of the servers in the cloud data center and gets allocated certain amount of CPU and memory resources in order to perform certain computing tasks.
- ▶ Each IoT application is hosted on a VM that has its own operating system (OS). The hypervisor (virtual machine manager) monitors those running VMs and manages how these VMs share the server's hardware.
- ▶ The hypervisor also provides the logical separation among the VMs and also separates each VM from the underlying hardware. The hypervisor has also a migration module that manages how to move a VM that is currently hosted on the server to another server. The migration module also manages the reception of a VM that is moved from other servers.

# Cloud Domain Attacks and Countermeasures

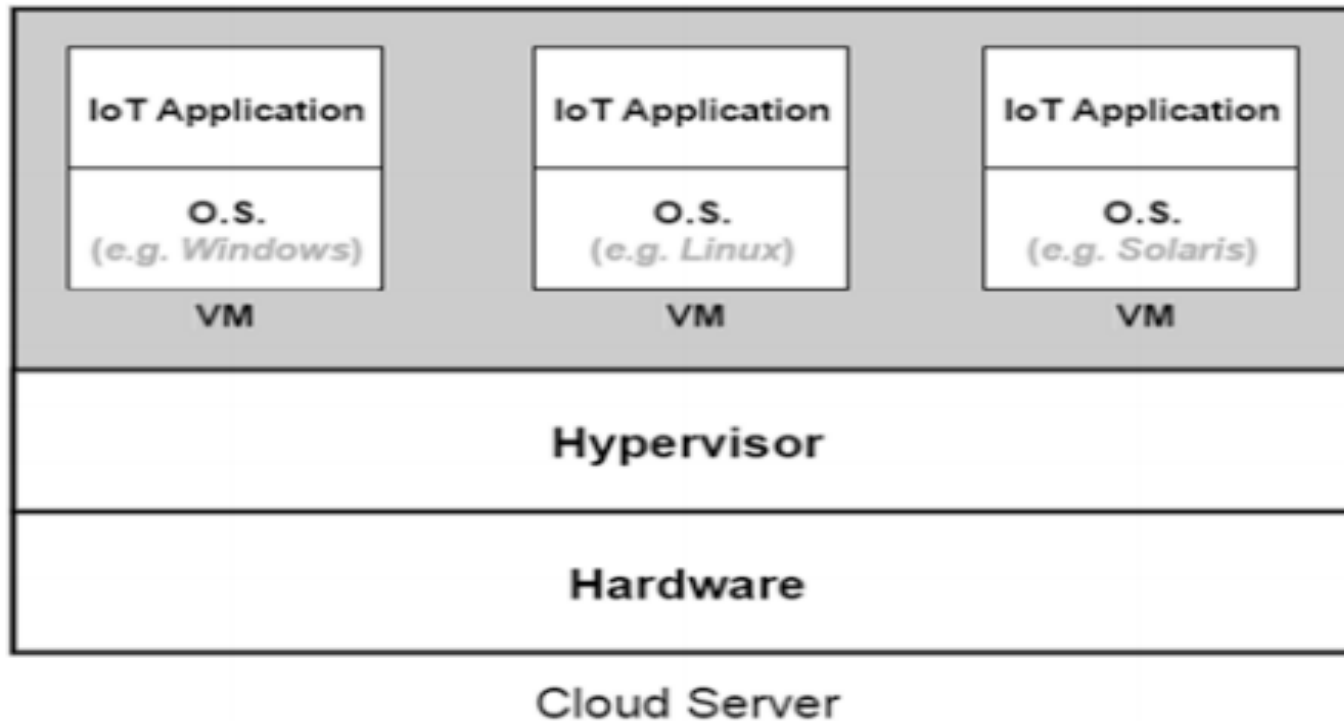


Fig. 8.3 Illustration of how multiple IoT applications can be hosted on the same server, thanks to virtualization

# Cloud Domain Attacks and Countermeasures

- ▶ The IoT applications running in the cloud domain are susceptible to numerous security attacks.

(a) **Hidden-Channel Attacks:** Although there is a logical separation among the VMs running on the same server, there are still some hardware components that are shared among those VMs such as the cache. This opens opportunities for data leakage across the VMs that reside on the same server. Three steps are followed by the attacker in order to leak information from a target VM.

Step1: **Mapping Target VM:** A VM in a cloud data center is to locate where the target VM resides. A cloud data center is typically divided into multiple management units called clusters, where each cluster is located in a certain geographical location and is made up of thousands of servers.

# Cloud Domain Attacks and Countermeasures

- ▶ The attacker can **infer based on the VM's external IP address** on what cluster the VM resides, as cloud clusters are usually placed in different geographical locations and have different IP addresses.
- ▶ Step2: **Malicious VM Placement**: having identified on what cluster and on what zone the target VM resides, the next step toward launching an attack against the target VM is to place a malicious VM on the same server where the target VM resides. In order to do that, **the attacker rents a VM in the same cluster as the target VM.**
- ▶ The cloud provider's scheduling algorithm places the rented VM on one of the servers within one of the cluster's zones. The attacker performs a traceroute from the rented VM to the target VM where the routing path that separates the rented VM and the target VM is identified. If the identified routing path shows multiple hops that separate the target VM and the rented VM, then the attacker knows that the rented VM was not placed on the same server as the target VM. The attacker then releases the rented VM and requests a new one.

# Cloud Domain Attacks and Countermeasures

- ▶ **Step3: Cross-VM Data Leakage:** Having placed a malicious VM on the same server as the target VM, the attacker now tries to learn some information about the target VM by exploiting the fact that although VMs are separated logically, thanks to virtualization, **they still share certain parts of the server's hardware such as the instruction cache and the data cache.** The attacker can now, for example, learn what lines of cache (data or instruction) **the target VM has accessed recently.**
- ▶ This gives information to the malicious VM about what addresses the target VM has accessed recently. Knowing what addresses the target VM accesses over time can help the malicious VM recover parts of the security keys that the target VM is using.
- ▶ Different counter measures can be taken to prevent hidden-channel attacks from taking place. **The first two steps needed to launch this attack can be prevented by not allowing the VMs hosted in the cloud data center to send probing packets such as traceroute packets.**

# Cloud Domain Attacks and Countermeasures

- ▶ **Hard Isolation:** The basic idea behind this preventive technique is to maintain high levels of isolation among the VMs. One way to do this is to separate the cache dedicated for each VM through hardware or software. Another way to achieve hard isolation is by assigning only one VM to each server. Although this completely prevents data leakages across VMs, it is not a practical solution as it leaves the servers within the cloud data center under utilized.
- ▶ **Cache Flushing:** This technique flushes the shared cache every time the allocation of the cache is switched from a VM to another. The downside of this countermeasure is that the VMs running on the server will experience frequent performance degradation as the shared cache will be emptied every time a switch from a VM to another occurs, which increases the time needed to access and fetch data.
- ▶ **Noisy Data Access Time:** This technique adds random noise to the amount of time needed to fetch data, which makes it hard to tell whether or not the data was fetched from the cache or from the memory. By doing this, it becomes harder for a malicious VM to identify what segments of the cache were populated by another VM that shares the same server.

# Cloud Domain Attacks and Countermeasures

- ▶ **Limiting Cache Switching Rate:** A mitigation technique to limit the amount of data that can be leaked across VMs can be achieved by **limiting how often the cache is switched from a VM to another**. The idea here is that if the cache is not switched from a VM to another too soon, then the content of the cache will be modified a lot by the VM that possesses the cache.
- ▶ **VM Migration Attacks:** The virtualization technology supports live VM migration, which allows moving a VM transparently from a server to another. The term live refers here to the fact that the application running on the VM is disrupted for a very short duration due to this migration where the disruption is as low as hundreds of milliseconds.



## The attacks that exploit VM migration

- ▶ The attacks that exploit VM migration can be divided into two subcategories based on the target plane:
- ▶ **Control Plane Attacks:** These attacks target the module that is responsible for handling the migration process on a server which is called the migration module that is found in the hypervisor. By exploiting a bug in the migration module software, the attacker can hack the server and take full control over the migration module.
- ▶ **Migration Flooding:** This attack is illustrated in Fig. 8.4 where the attacker moves all the VMs that are hosted on the hacked server to a victim server that does not have enough resource capacity to host all the moved VMs.

# the migration flooding attack

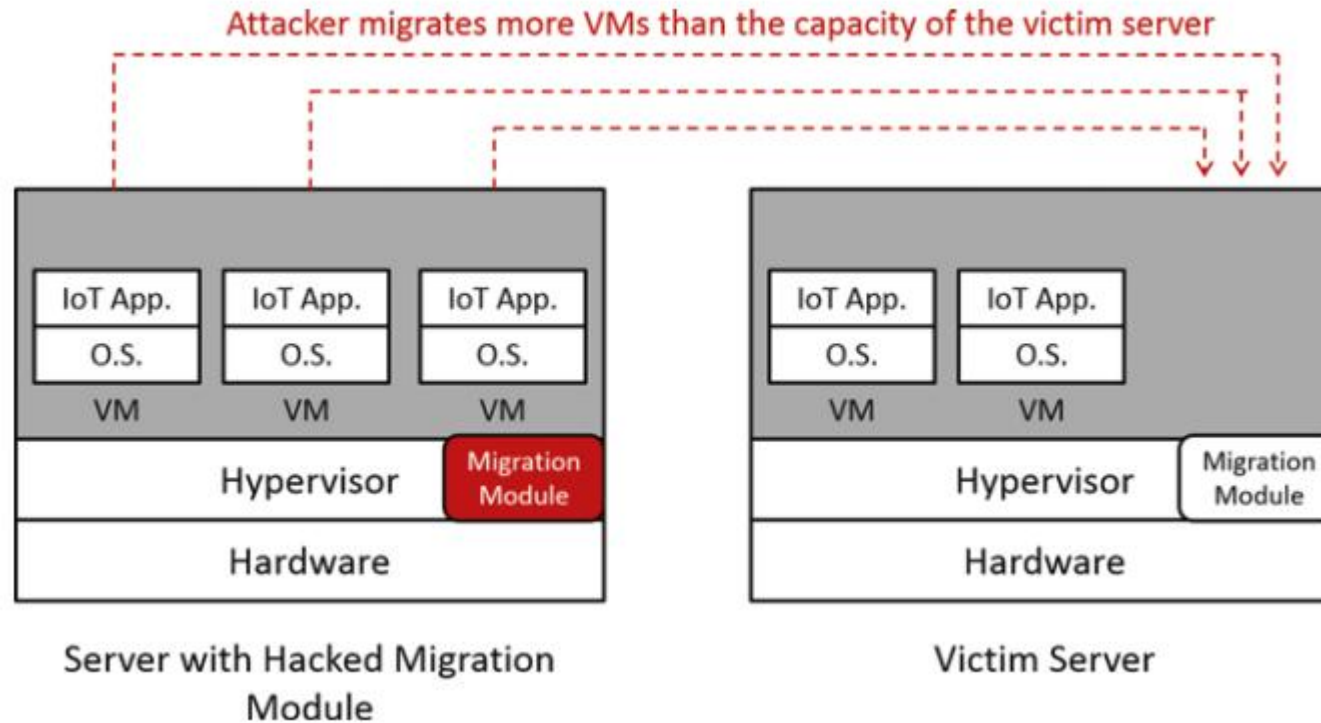


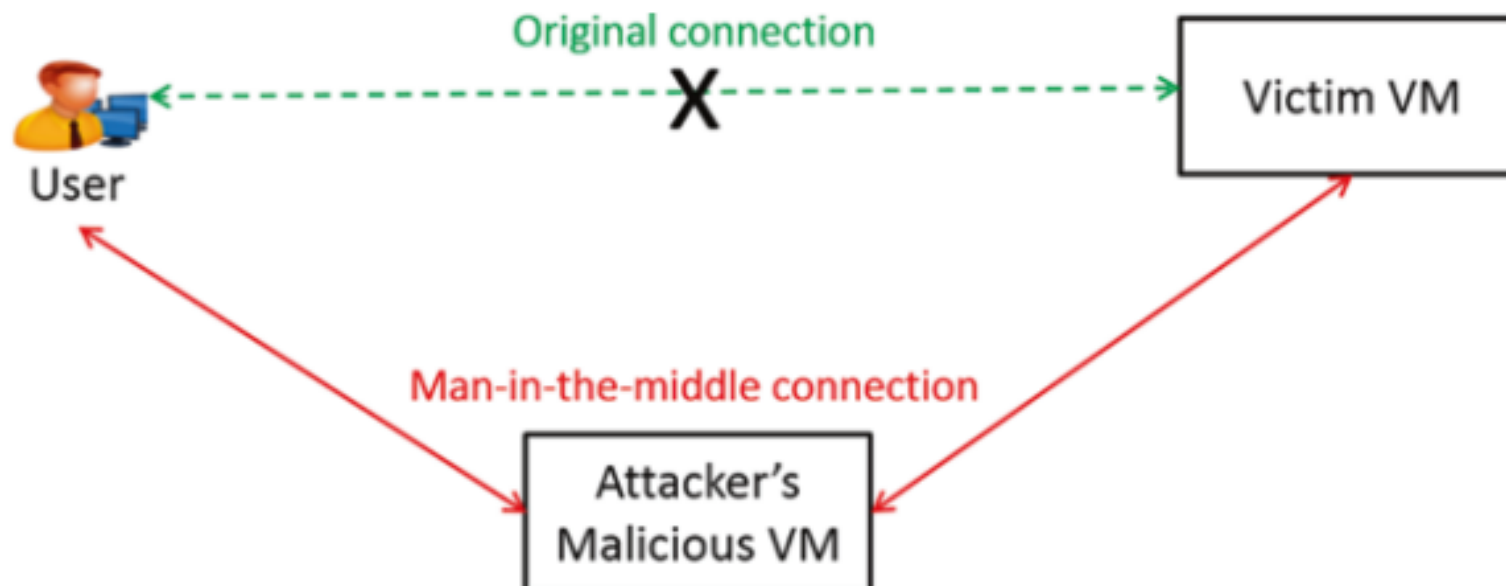
Fig. 8.4 Illustration of the migration flooding attack

# The migration flooding attack

- ▶ **False Resource Advertising:** The hacked server claims that it has a large resource slack. This attracts other servers to off-load some of their VMs to the hacked server so that the cloud workload gets distributed over the cloud servers. After moving VMs from other servers to the hacked server, the attacker can exploit other vulnerabilities to break into the offloaded VMs.
- ▶ **Data Plane Attacks:** These constitute the second type of VM migration attacks, and those attacks target the network links over which the VM is moved from a server to another. Examples of data plane attacks include:

# Man-in-the-middle attack

- ▶ **Sniffing Attack:** where an attacker sniffs the packets that are exchanged between the source and destination and reads the migrated memory pages.
- ▶ **Man-in-the-Middle Attack:** the attacker fabricates a gratuitous (address resolution protocol that translate between layer 2 mac address and layer 3 IP addresses) ARP reply packet similar to the one that is usually sent when a VM moves from a server to another.



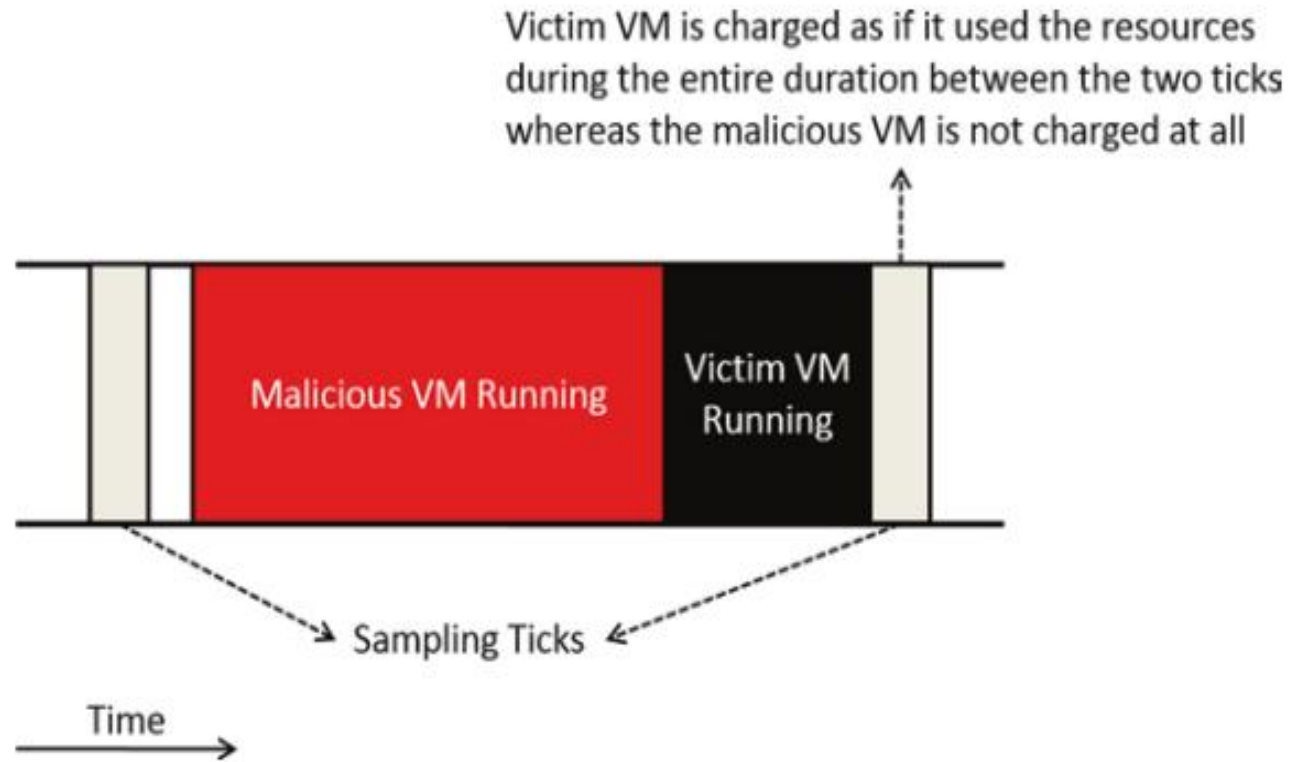
# Man-in-the-middle attack

- ▶ This fabricated ARP packet informs the routing devices that the physical address where the victim VM resides was changed to become the physical address of the attacker's malicious VM. Now the incoming packets that are **destined to the victim get routed to the new physical address where the attacker resides**. The attacker can then passively monitor the received packets while continuing to forward them to the actual physical address where the victim VM resides so that the victim does not detect that any malicious activity is going on.
- ▶ In order to secure VM migration, **mutual authentication should be performed** between the server initiating the migration and the server that will be hosting the migrated VM. The control messages that are exchanged between the servers **to manage the migration should also be encrypted and signed by the entity that is generating those control messages in order to avoid altering the content**.

# Man-in-the-middle attack

- ▶ (c) Theft-of-Service Attack: In this attack a malicious VM misbehaves in a way that makes the hypervisor assign to it more resources than the share it is supposed to obtain. This extra allocation of resources for the malicious VM comes at the expense of the other VMs that share the same server as the malicious VM, where these victim VMs get allocated less share of resources than what they should actually obtain, which in turn degrades their performance.

# Cloud Domain Attacks and countermeasures



The fact that Xen performs periodic sampling can be exploited by a malicious VM by using one of the cores at times other than the sampling time. As illustrated in Fig

Fig. Illustration of the theft-of-service attack

# countermeasures

- ▶ Two countermeasures were proposed to handle this attack.
- ▶ (1)The first countermeasure is to **log more accurately the start and end time** when each VM was utilizing the cores using accurate clocks.
- ▶ (2)Another solution is to **randomize the sampling times**



# Other attacks

- ▶ **(d) VM Escape Attack:** Virtual machines are designed in a way that isolate each VM from the other VMs running on the same server, which prevents VMs from accessing data that belongs to other VMs that reside on the same server. However, in reality software bugs can be exploited to break this isolation. If a VM escapes the hypervisor layer and reaches the server's hardware, then the malicious VM can gain root access to the whole server where it resides.
- ▶ **(e) Insider Attacks:** However, some sensitive applications may have serious concerns about hosting their collected information on the cloud data center in the first place as the cloud data center administrators will in that case have the ability to access and modify the collected data. Different techniques were proposed to protect the data from these insider attacks.

# Fog Domain Attacks and Countermeasures

- ▶ The fog device performs different operations on the collected data which include data aggregation, data preprocessing, and data storage. The fog device may also perform some reasoning operations on the collected data. After processing and aggregating the collected data, the fog device forwards these data to the cloud domain.
- ▶ Not only fog devices are connected with the cloud domain, but also fog devices are usually connected with each other in order to allow the fog devices connecting different smart objects to communicate directly with each other and in order to coordinate assigning objects to fog devices as their location changes.

# Summary of the security attacks in the cloud domain

| Attack                  | Vulnerability reason                                                                                           | Security violation                           | Countermeasures                                                                                      |
|-------------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------|------------------------------------------------------------------------------------------------------|
| Hidden-channel attack   | Shared hardware components (e.g., cache) among the server's VMs                                                | Confidentiality                              | Hard isolation<br>Cache flushing<br>Noisy data access time<br>Limiting cache switching rate          |
| VM migration attacks    | VM migration software bugs<br>VM migration is performed without authentication<br>Memory pages copied in clear | Confidentiality<br>Integrity<br>Availability | Server authentication<br>Encrypting migrated memory pages                                            |
| Theft-of-service attack | Periodic sampling of VMs' used resources                                                                       | Availability<br>Non-repudiation              | Fine-grain sampling using high precision clocks<br>Random sampling                                   |
| VM escape attack        | Hypervisor software bugs                                                                                       | Confidentiality<br>Availability<br>Integrity | Add an isolation domain between the hypervisor and hardware                                          |
| Insider attacks         | Lack of trust in cloud administrators                                                                          | Confidentiality<br>Integrity                 | Homomorphic encryption<br>Secret storage through data chopping and permutation based on a secret key |

There are three key differences that distinguish fog devices from cloud servers:

- ▶ **1. Location:** fog devices are placed close to the smart objects. This placement plays an important role in giving the fog devices the ability to respond quickly to changes in the reported data.
- ▶ **2. Mobility:** Since the location of the smart object may change over time, then the VMs created to handle those objects at the fog domain must be moved from a fog device into another, in order to keep the processing that is performed in the fog device close to the object that is generating data.
- ▶ **3. Lower Computing Capacity:** The fog devices that are installed in a certain location are expected to have a lower computing capacity when compared to capacities offered by cloud data centers

# The security threats that are specific to the fog domain

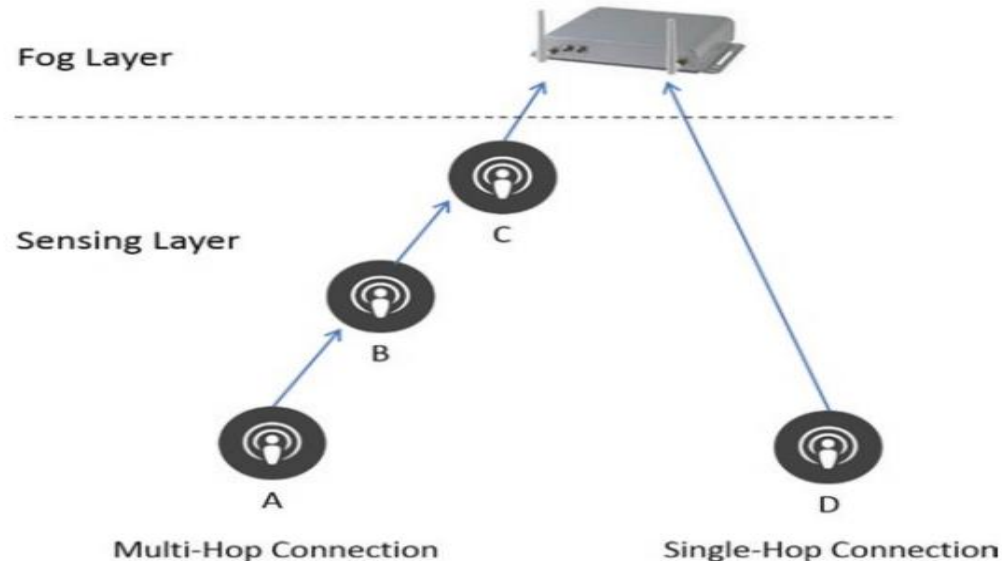
- ▶ **Authentication and Trust Issues:** The fact that fog devices don't require a large facility space or a high number of servers compared to cloud data centers will encourage many small and less-known companies to install virtualized fog devices in dense areas. Unlike cloud data centers which are offered by well-known companies, fog devices are expected to be owned by multiple and less-known entities.
- ▶ **Higher Migration Security Risks:** Although VM migration is common in both the cloud and the fog domains, there is an important difference between the migration in the cloud domain and that in the fog domain. While the migrated VMs in the cloud domain are carried over the cloud data center's internal network, the migrations from a fog device into another are carried over the Internet.

# The security threats that are specific to the fog domain

- ▶ **Higher Vulnerability to DoS Attacks:** Since fog devices have lower computing capacities, this makes them a low-hanging-fruit for denial of service (DoS) attacks where attackers can easily overwhelm fog devices when compared to the cloud data centers, where a huge number of servers that have high computing capacity are available.
- ▶ **Additional Security Threats Due to Container Usage:** In order to provide the computing needs for a larger number of connected objects, the fog device may use containers rather than VMs to allocate the resource demands for each connected object. The main difference between a container-based virtualization and full virtualization is the fact that **containers share not only the same hardware but also the same operating system** with the other containers that are hosted on the same fog device.
- ▶ **Privacy Issues:** The fog device can infer the location of all the connected smart objects. This allows the **fog device to track users or to know their commuting habits which may break the privacy** of the users carrying those objects. New mechanisms should be developed in order to make it harder for fog devices to track the location of the smart objects over time.

# Sensing Domain Attacks and Countermeasures

- ▶ We summarize next some of the most well-known ones:
- ▶ (a) Jamming Attack: This attack causes a service disruption and takes one of two forms:
- ▶ 1. **Jamming the Receiver:** This attack targets the physical domain in the OSI stack of the receiver where a malicious user (called the jammer) emits a signal (called the jamming signal) that interferes with the legitimate signals that are received at the receiver side.



Multi-hop versus direct connection between the smart object and the fog device

# Sensing Domain Attacks and Countermeasures

- ▶ 2. **Jamming the Sender**: Unlike the previous attack, this type targets the data link layer at the OSI layer of the sending object where the jammer in this attack sends a jamming signal that prevents the neighboring objects from transmitting their packets as they sense the wireless channel to be busy and back off waiting for the channel to become idle.



## Different jamming strategies that a jammer may follow to launch a jamming attack.

- ▶ **Constant Jamming:** The attacker continuously transmits a random jamming signal all the time. The main limitation of this attack is that it can be detected easily by observing random bits that don't follow the pattern dictated by the MAC protocol.
- ▶ **Deceptive Jamming:** This is similar to the constant jamming with the exception that the jammer conceals its malicious behavior by transmitting legitimate packets that follow the structure of the MAC protocol rather than sending random bits.

# Different jamming strategies that a jammer may follow to launch a jamming attack.

- ▶ **Reactive Jamming:** This is a strategy for jamming the receiver that is suitable for the case when the jamming device has a limited power budget. The jammer in that case listens to the medium and transmits a jamming signal only after it senses that a legitimate signal is being transmitted in the medium. This is more power efficient than continuously transmitting signals as listening to the channel consumes less power than transmitting signals.
- ▶ **Random Jamming:** The jammer alternates between sending a jamming signal and remaining idle for random periods of time in order to hide the malicious activity.

# Address jamming attacks

- ▶ **Frequency Hopping:** This is a preventive technique where the sender and receiver switch from a frequency to another in order to escape from any possible jamming signal. Switching from a frequency to another is based on a generated random sequence that is known only for the sender and receiver.
- ▶ **Spread Spectrum:** This technique uses a hopping sequence that converts the narrow band signal into a signal with a very wide band, which makes it harder for malicious users to detect or jam the resulting signal.

# Address jamming attacks

- ▶ **Directional Antennas:** The use of directional antennas can mitigate jamming attacks from being successful as the sender and receiver antennas will have less sensitivity to the noise coming from the random directions that are different from the direction that connects the sender and the receiver.
- ▶ **Jamming Detection:** Different detective techniques were proposed in the literature to detect jamming attacks. The receiver can detect that it is a victim of a jamming attack by collecting features such as the received signal strength (RSS)
- ▶ **Vampire Attack:** This attack exploits the fact that the majority of IoT objects have a limited battery lifetime where a malicious user misbehaves in a way that makes devices consume extra amounts of power so that they run out of battery earlier thereby causing a service disruption.

# Four types of vampire attacks based on the strategy used to drain power:

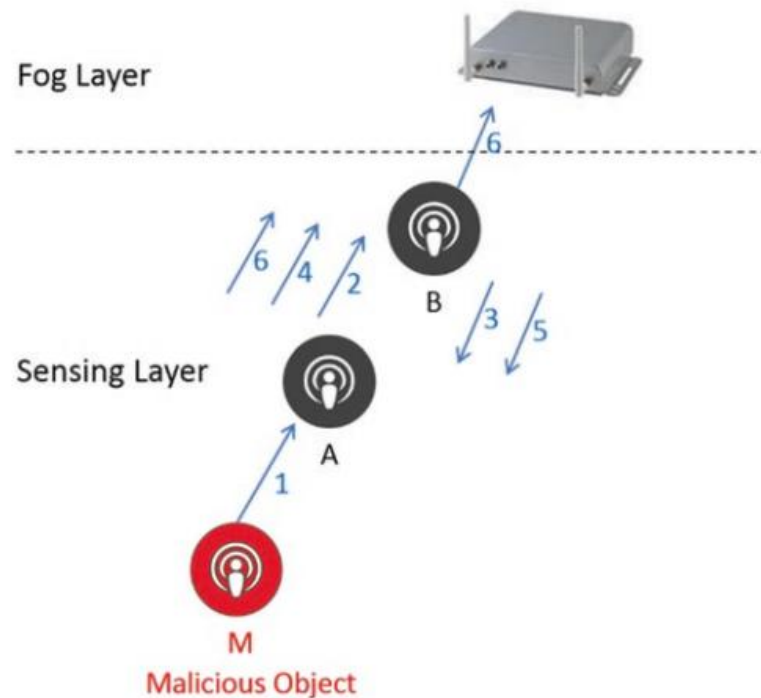
- ▶ **Denial of Sleep:** Different data link layer protocols were proposed to reduce the power consumption of smart objects by switching them into sleep whenever they are not needed. Examples of these protocols include S-MAC and T-MAC protocols.
- ▶ **Flooding Attack:** The adversary can flood the neighboring nodes with dummy packets and request them to deliver those packets to the fog device, where devices waste energy receiving and transmitting those dummy packets.

# Four types of vampire attacks based on the strategy used to drain power:

- ▶ **Carrousel Attack:** This attack targets the network layer in the OSI stack and can be launched if the routing protocol supports source routing, where the object generating the packets can specify the whole routing path of the packets it wishes to send to the fog device.
- ▶ **Stretch Attack:** This attack also targets the network layer in the OSI stack. If the routing protocol supports source routing, then a malicious object can send the packets that it is supposed to report to the fog device through very long paths rather than the direct and short ones as illustrated in Fig. 8.8

# Stretch Attack:

- ▶ This attack also targets the network layer in the OSI stack. If the routing protocol supports source routing, then a malicious object can send the



packets that it is supposed to report to the fog device through very long paths rather than the direct and short ones as illustrated in Fig. 8.8. Even if source routing is not supported, the attacker can select a next hop that does not have the shortest path to the fog device in order to increase the power consumption of the objects that will be responsible to deliver those packets

Fig. 8.8 Illustration of the carousel attack where the numbered arrows show the path specified by the malicious objects that the packets generated by the malicious object follow

# Selective-Forwarding Attack:

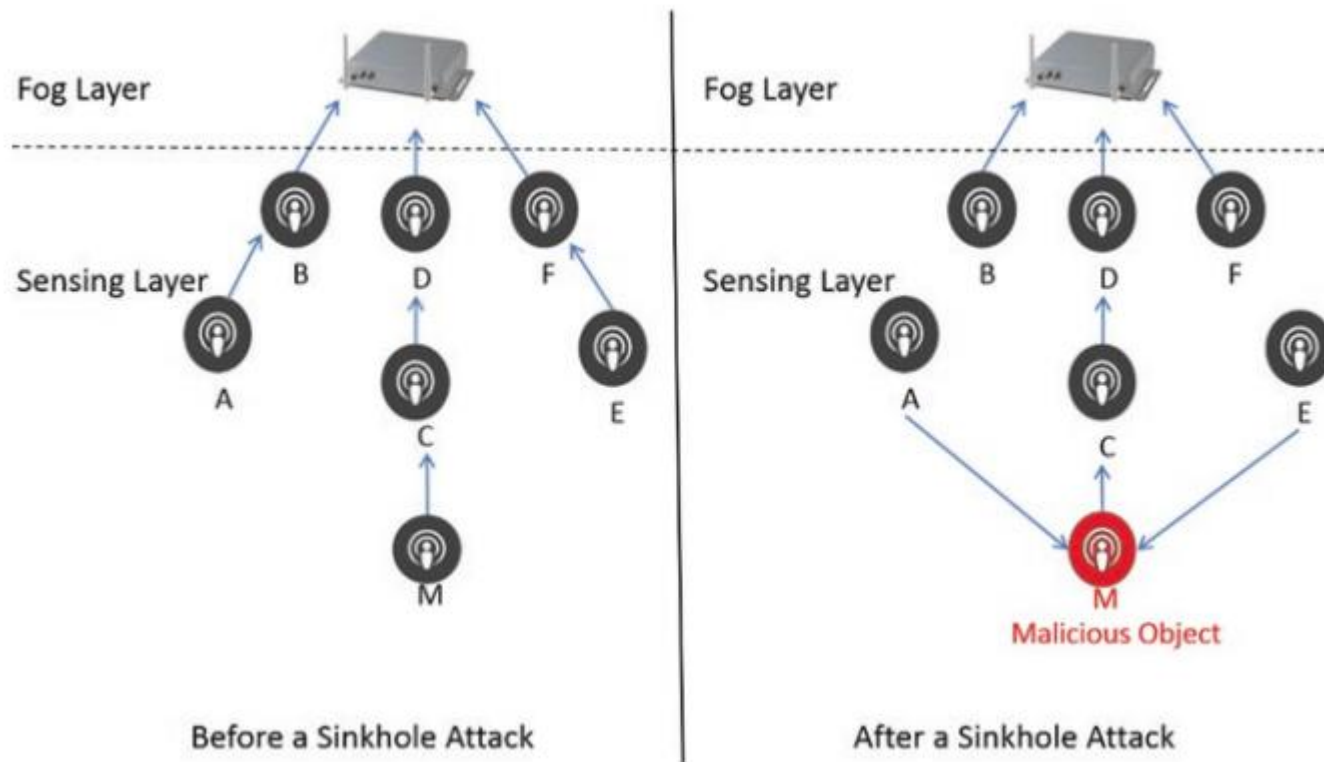
- ▶ This attack takes place in the case when the object can't send its generated packets directly to the fog device but must rely on other objects that lie along the path toward the fog device to deliver those packets .
- ▶ A malicious object in this attack does not forward a portion of the packets that it receives from the neighboring objects. A special case of this attack is the lackhole attack where the attacker drops the entire set of packets that it receives from the neighboring objects. The best way to prevent packet drops from taking place for sensitive IoT applications is to increase the transmission capability of the objects so that they can reach the fog device directly without the need for help from intermediate objects.
- ▶ Path redundancy is one of those solutions, where each object forwards each generated packet to multiple neighboring objects, where multiple copies of the same packet get delivered to the fog device through different paths.



# Sinkhole Attack

- ▶ A malicious object claims that it has the shortest path to the fog device which attracts all neighboring objects that don't have the transmission capability to reach the fog device to forward their packets to that malicious object and count on that object to deliver their packets.
- ▶ Now all the packets that are originating from the neighboring nodes pass by this malicious node. This gives the malicious node the ability to look at the content of all the forwarded packets if data is sent with no encryption. Furthermore, the malicious object can drop some or all of the received packets as we explained previously in the selective-forwarding attack.

# Sinkhole Attack



Network topology before and after a sinkhole attack. The malicious object M claims that it has a shorter route to reach the fog device which attracts the neighboring objects A and E to rely on M to deliver their packets

## Summary of the security attacks targeting the sensing domain

| Attack                      | Target OSI layer      | Vulnerability reason            | Security violation              | Countermeasures                                                                                                                                                       |
|-----------------------------|-----------------------|---------------------------------|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Jamming attack              | Physical<br>Data link | Shared wireless channel         | Availability                    | Frequency hopping<br>Spread spectrum<br>Directional antennas<br>Jamming detection techniques                                                                          |
| Vampire attack              | Data link<br>Network  | Limited battery lifetime        | Availability<br>Freshness       | Rate limitation<br>Drop packets with a source route that contains a loop<br>Monitor whether or not the forwarded packets are making progress toward their destination |
| Selective-forwarding attack | Network               | Limited transmission capability | Availability                    | Increase transmission range<br>Path redundancy<br>Choose certain intermediate objects as checkpoints to acknowledge received packets                                  |
| Sinkhole attack             | Network               | Limited transmission capability | Confidentiality<br>Availability | Analyze the collected routing information from multiple objects                                                                                                       |

# Summary

- ▶ This chapter analyzed IoT from a security and privacy perspectives. Ignoring security and privacy will limit the applicability of IoT and will have serious results on the different aspects of our lives given that all the physical objects in our surrounding will be connected to the network. In this chapter, the IoT security challenges and IoT security requirements were identified. A three-domain IoT architecture was considered in our analysis where we analyzed the attacks targeting the cloud domain, the fog domain, and the sensing domain. Our analysis describes how the different attacks at each domain work and what defensive countermeasures can be applied to prevent, detect, or mitigate those attacks. We hope that security and privacy will be considered at the early design stage of IoT in order to avoid the common pitfall of considering security as an after thought.

# References

- ▶ 1. Book - Internet of Things from hype to reality  
the road to digitization (Chapter 8)  
By Ammar Rayes and Samer Salam

# Question



Thank you

